BF25.4

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Background

Purpose:
This lab will help us determine the relationship between voltage, current, and resistance in a circuit
Hypothesis:
The correct relationship between V, I, and R in a circuit is $V = IR$. Voltage is proportional to both current and resistance.
Variables:
Independent:
- Resistance of the circuit
- Voltage of the circuit
Dependent:
- Current of a circuit
Constants:
- Type of batteries
- Type of lightbulb (resistor)
- Amount of wire used
Justification for hypothesis:
Intuitively, the more the flow of electrons is impeded in a circuit, the more electrons will travel through the circuit. The specific indirect relationship between current and resistance is not guaranteed to be a first order indirect relationship, but intuitively, the percent increase of resistance should equal the percent increase of resistance, because any other relationship with resistance would make less sense as to why a different power of the percent increase of resistance should change the percent increase of resistance. Regardless, in practice, there are so many other factors like the internal resistance of the batteries, the resistance of the wires, and the fact that real resistors are not perfectly Ohmic, so I'm interested as to how different from Ohmic our results will be.

Materials

- 3 batteries
- 2 lightbulbs
- 2 alligator clips
- Voltmeter

Procedure

Part 1 - Changing voltage

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1. Connect one battery to an alligator clip on either end, and then connect each of those clips to either end of a lightbulb.
2. Record the current across the lightbulb using a voltmeter.
3. Disconnect the circuit, and reconnect it to count as another trial, recording with the voltmeter each time. Repeat for 3 trials.
4. Add another battery to the circuit right next to the original battery without adding more alligator clips. Repeat 2-3 for 3 trials with 2 batteries.
5. Repeat 4 with 3 batteries.

Part 2 - Changing resistance

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1. Connect one battery to an alligator clip on either end, and then connect each of those clips to either end of a lightbulb.
2. Record the current across the lightbulb using a voltmeter.
3. Disconnect the circuit, and reconnect it to count as another trial, recording with the voltmeter each time. Repeat for 3 trials.
4. Add another lightbulb to the circuit right next to the original lightbulb without adding more alligator clips. Repeat 2-3 for 3 trials.

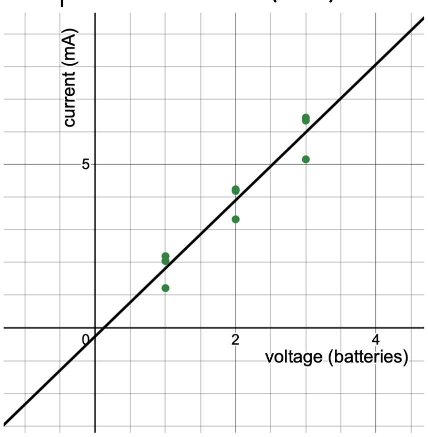
Data and Results Tables

Part 1 - Changing voltage

Constant one lightbulb

Current (mA)	Trial 1	Trial 2	Trial 3	Average
1 battery	1.22	2.05	2.20	1.82
2 batteries	4.25	4.20	3.33	3.93
3 batteries	6.36	6.45	5.17	5.99

Graph of Current (mA) vs Voltage (batteries)



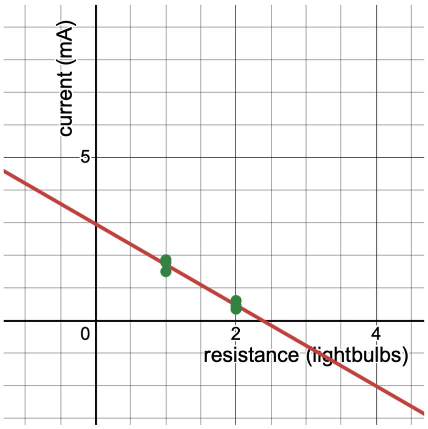
$y = 2.085x - 0.255556$

Part 2 - Changing resistance

Constant one battery

Current (mA)	Trial 1	Trial 2	Trial 3	Average
1 lightbulb	1.50	1.86	1.78	1.71
2 lightbulbs	0.45	0.35	0.61	0.47

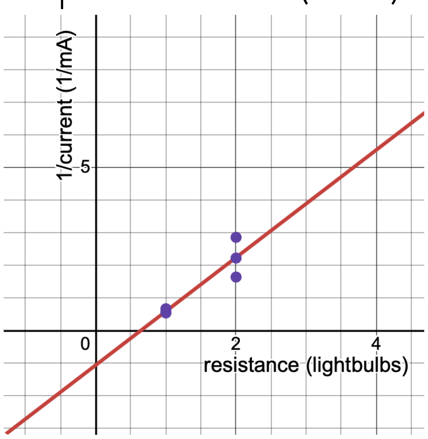
Graph of Current (mA) vs Resistance (lightbulbs)



$y = -1.24333x + 2.95667$

1/Current (1/mA)	Trial 1	Trial 2	Trial 3	Average
1 lightbulb	0.67	0.54	0.56	0.58
2 lightbulbs	2.22	2.86	1.64	2.13

Graph of 1/Current (1/mA) vs Resistance (lightbulbs)



$y = 1.65087x - 1.06217$

Analysis

Based off the graphs, it is clear there is a linear relationship between V and I based off the first graph. Comparing the graphs of resistance and current versus the graph of resistance and 1/current, it is more apparent that there is a linear relationship with 1/current because the slope is positive and goes approximately through the origin. This section of the analysis proves Ohm's Law, because it confirms that $R \propto \frac{1}{I}$ and $V \propto R$. Combining these gives $V \propto IR$. Since resistance is a derived quantity from this specific equation, we can turn the proportionality into an equality symbol ($V = IR$).

The next part of this analysis attempts to estimate the resistance of the ligthbulb. Assuming that each battery is exactly $1.5V$, we can use the slope of the current vs voltage line to find the resistance of the lightbulb. Since $V = IR$, $R = \frac{V}{I}$. Since $slope = \frac{y}{x} = \frac{I}{V}$, the resistance of the lightbulb is the reciprocal of the slope. The slope is $2.085\frac{mA}{battery}$. Doing dimensional analysis, we can see that the resistance of the lightbulb is $\frac{1battery}{2.085mA} * \frac{1.5V}{1battery} * \frac{1000mA}{1A} = \boxed{719\Omega}$.

- **Focused Analysis Questions:**
  - If a voltage-current graph were created using the data collected from an Ohm's law lab, what would the significance of the slope of the voltage-current graph for a circuit, and how does this relate to Ohm's law?
    - As I demonstrated above, the slope of the voltage current graph shows the resistance of the circuit. This relates to Ohm's Law because we can rewrite Ohm's Law as $R = \frac{V}{I}$.
  - Is the relationship between voltage and current data from the always linear, as predicted by Ohm's law? Why or why not?
    - The relationship between voltage and current is ideally linear, which is predicted by Ohm's Law. However, in reality, no voltage current graph will ever be perfectly linear because this assumes that the circuit is Ohmic, which is just an approximation of reality.

Conclusion

This lab supports the hypothesis that $V = IR$ as stated by Ohm's Law. I used linear regression to show the relationship between current and voltage, and the relationship between current and resistance. I calculated that the resistance of the lightbulb was about $719\ \Omega$ by assuming the circuit was Ohmic and that every battery was exactly $1.5V$. If I were to do this lab again, I would use different circuit layouts to get more values for equivalent resistance so that I could plot more points on my resistance 1/current graph.